

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1520

Gray'S Flat 230 kV
August 30, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1520. The Interconnection Customer proposes a 44.6 MW battery energy storage facility interconnecting at the Gray'S Flat 230 kV substation in the Ohio area. The request was received on July 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 546 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,292,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Fiddymment / Miami 100.0 kV line
Contingency: Queen Creek / Baraga Vlg 500.0 kV line
Loading: 121.7% (Rating: 156 MVA)
DFAX: 5.5%
Pre-project loading: 120.2%
2. Facility: Fiddymment 100/1kV Transformer Bank
Contingency: Baraga Vlg / Queen Crk 500.0 kV
Loading: 120.7% (Rating: 162 MVA)
DFAX: 6.8%
Pre-project loading: 118.8%
3. Facility: Minco / Barrett 230.0 kV
Contingency: Norwich / Barrett 500.0 kV
Loading: 109.8% (Rating: 723 MVA)
DFAX: 6.9%
Pre-project loading: 109.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fiddymment / Miami 100.0 kV Line	Rebuild line with higher-capacity conductor	\$3,993,000	23% (\$930,000)
Fiddymment 100/1 kV TX	Replace transformer with higher-rated unit	\$3,261,000	10% (\$322,000)
Barrett / Minco 230.0 kV	Reconductor line	\$5,874,000	1% (\$40,000)
	Total allocated to project		\$1,292,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.